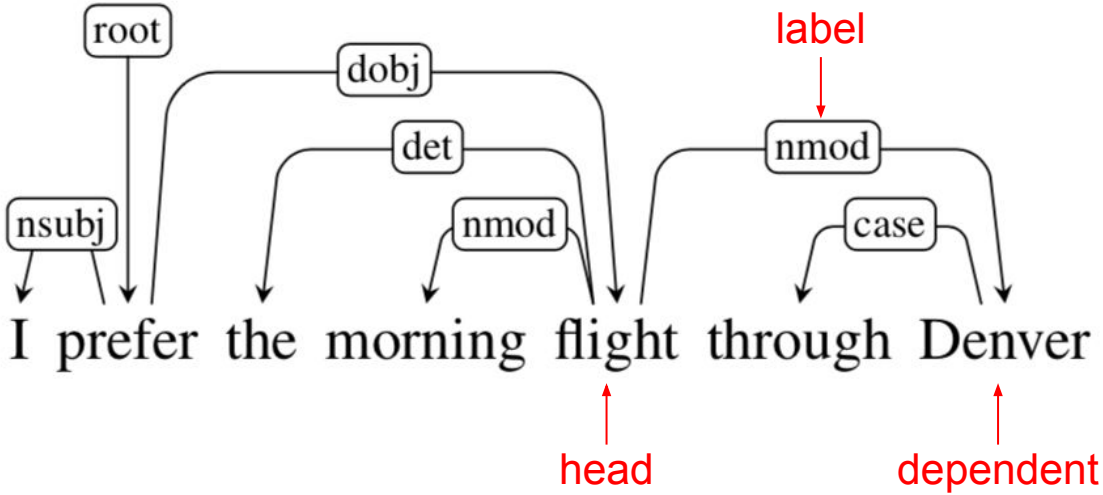
Evaluating Dependency Parsing Architectures on High and Low-Resource Languages

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Princeton University

Dependency Parsing

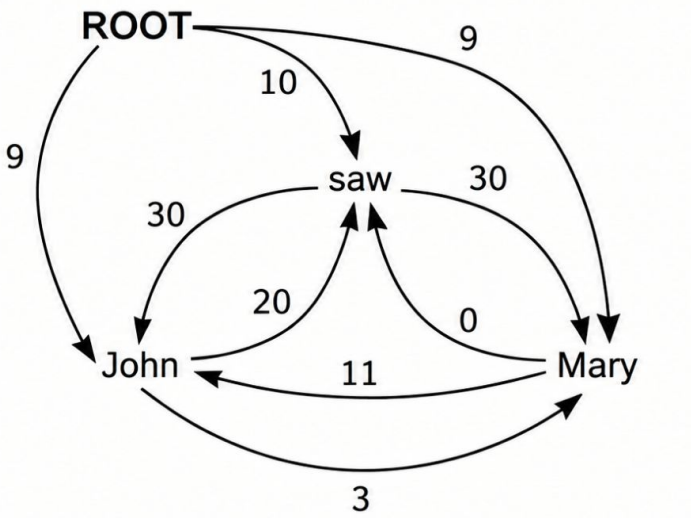


UAS: percentage of words with correct heads

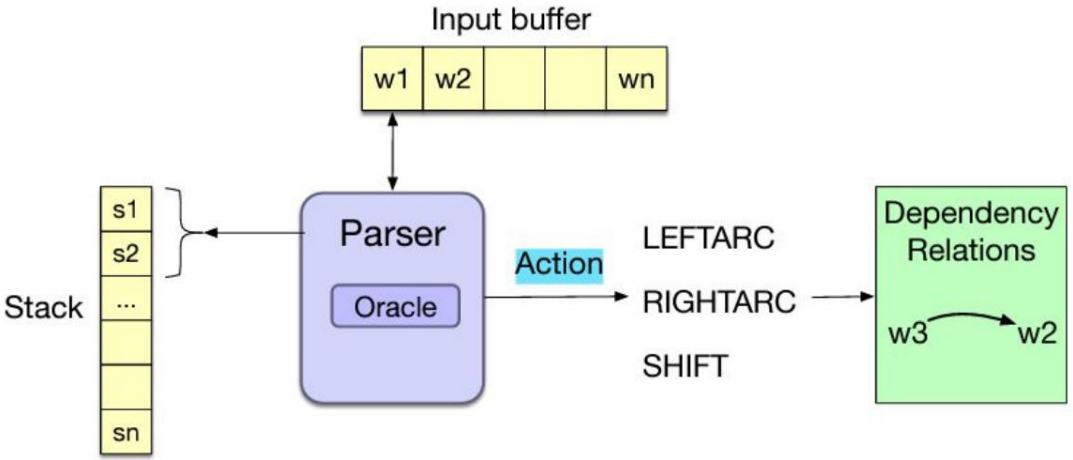
LAS: percentage of words with correct heads and labels

Jurafsky & Martin (2026), *Speech and Language Processing*, 3rd ed.

Neural Parsers



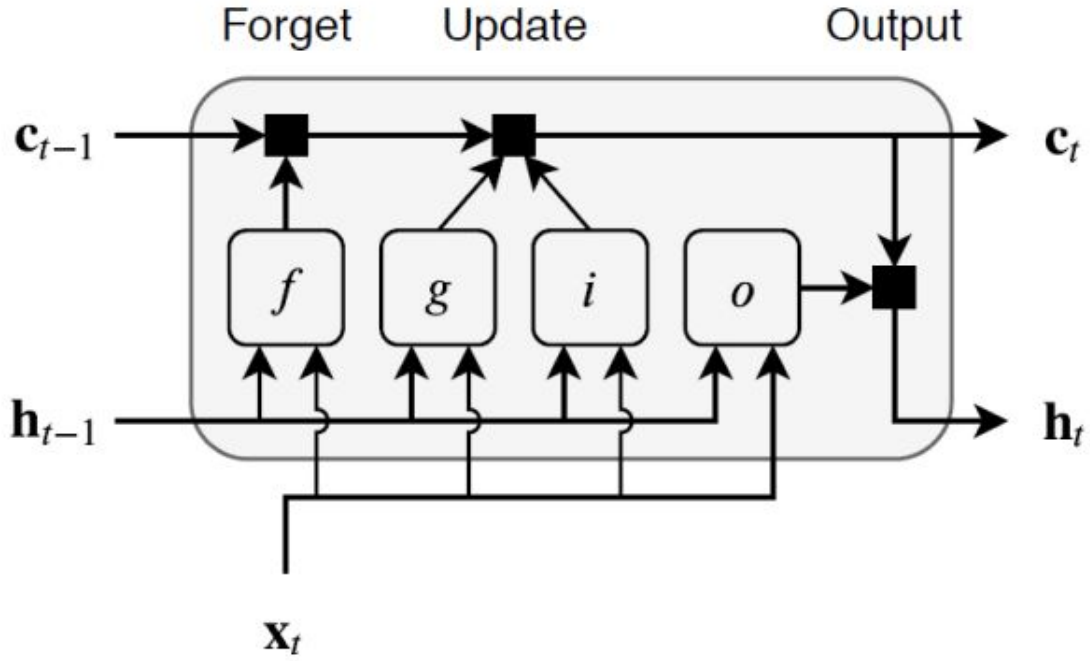
Graph-Based



Transition-Based

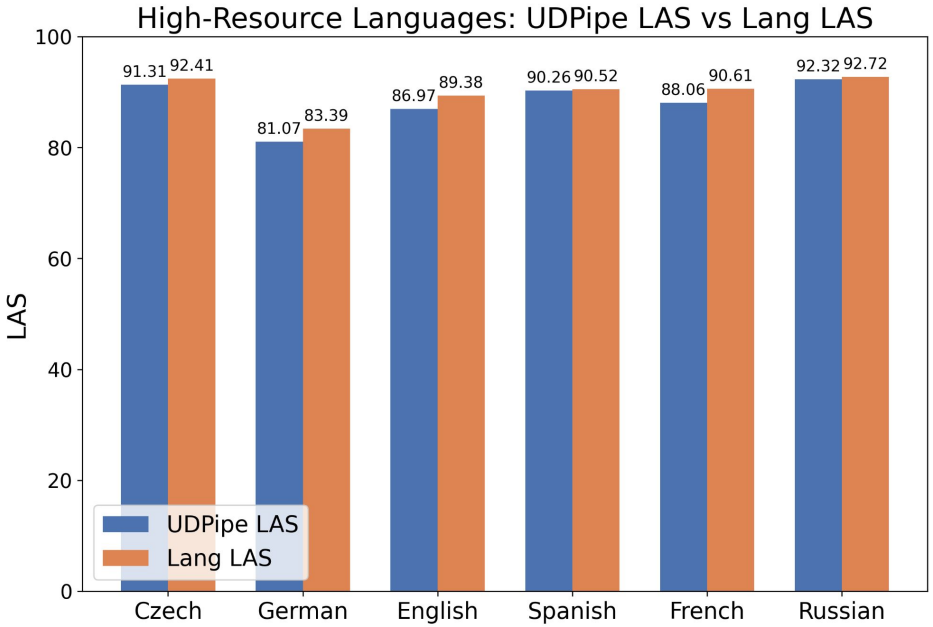
Toleu, A., Tolegen, G., & Mussabayev, R. (2019). *Comparison of Various Approaches for Dependency Parsing.* OPCS 2019, 192–195 12

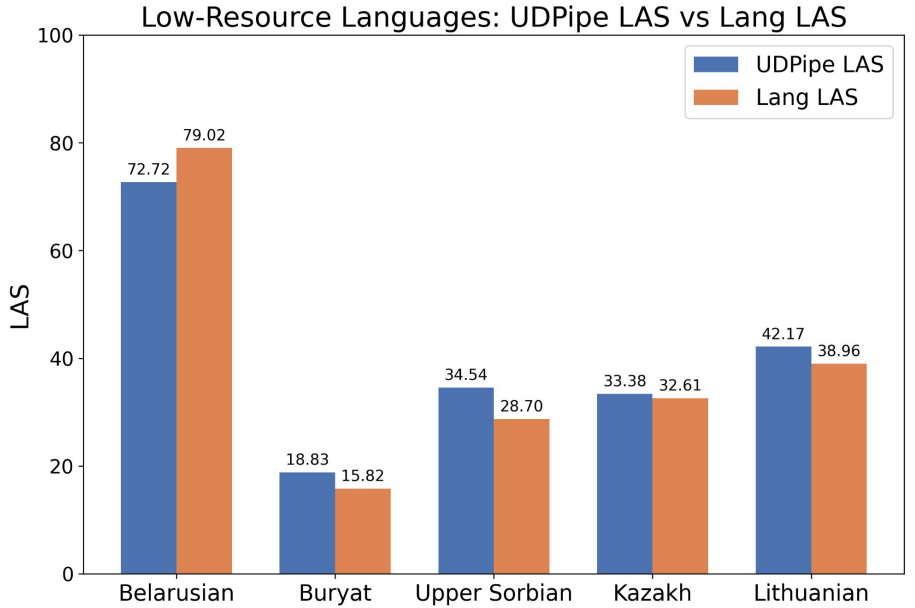
Long Short-Term Memory (LSTM)



MathWorks. (2026). *What Is Long Short-Term Memory (LSTM)?* MathWorks.

Pretrained Transformers





Adapted from Kondratyuk, D. & Straka, M. (2019). *75 Languages, 1 Model: Parsing Universal Dependencies Universally*. EMNLP-IJCNLP, 2779–2795.

Why?

1. Generalization
2. Inductive bias
3. Stability

Q: How well do different dependency parsing architectures generalize across high- and low-resource languages?

Architectures

LSTM-based Approaches

1. Biaffine (Dozat and Manning 2017)
2. Stack-Pointer (Ma et al. 2018)

Transformer-based Approaches

1. RemBERT
2. AfroXLMR-large

Languages

Resource	Language	Dataset	Training Size
High	French	French GSD	10,000*
	Dutch	Alpino	10,000*
	Romanian	Romanian RRT	7,619
	Turkish	Turkish IMST	4,508
Low	Afrikaans	AfriBooms	1,547
	Swahili	AfriSUD	1,187
	Kinyarwanda		1,098
	Nigerian Pidgin		1,085
	Wolof		736
	Yoruba		694
	Hausa		690
	Xhosa		369

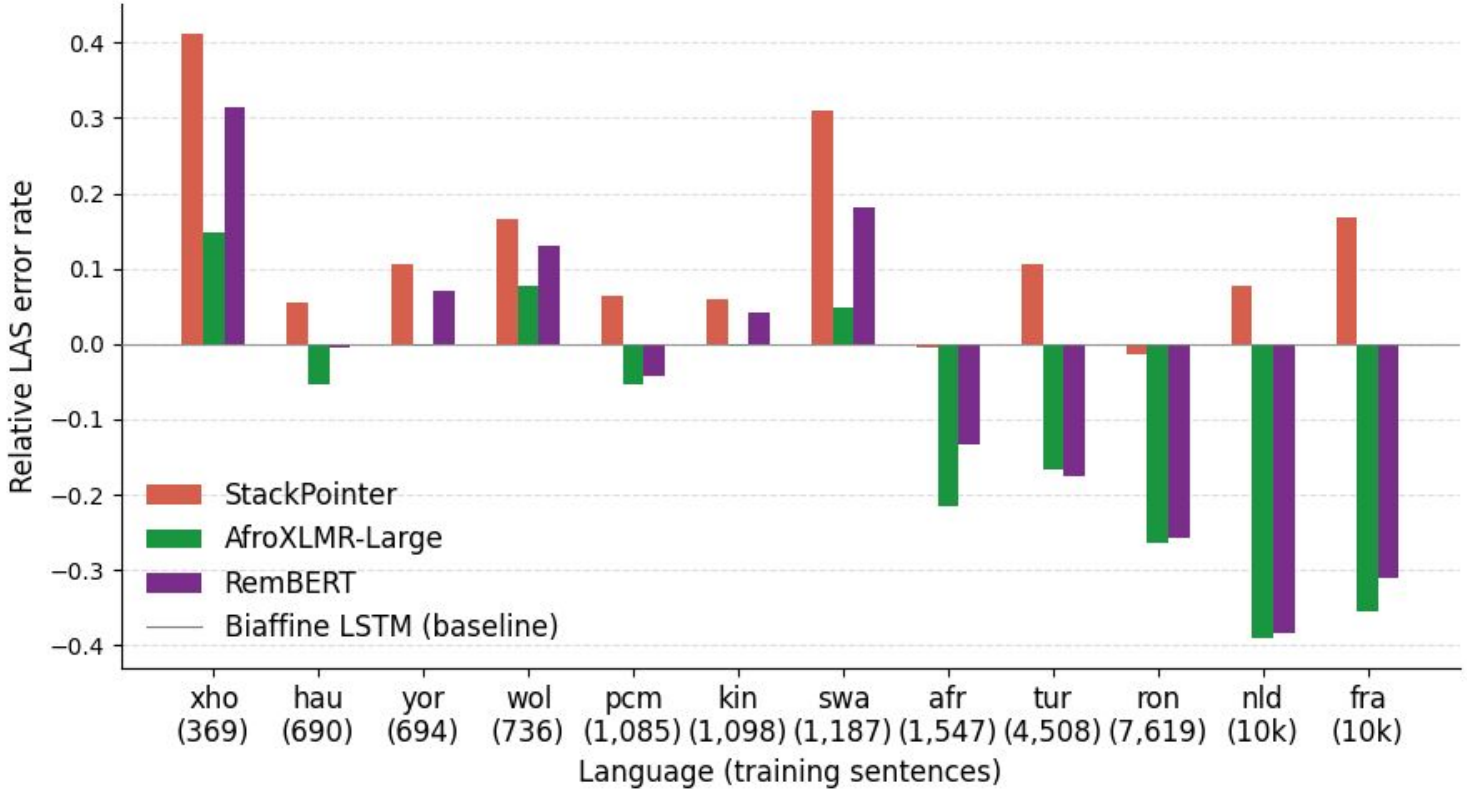
*Downsampled

Relative Error

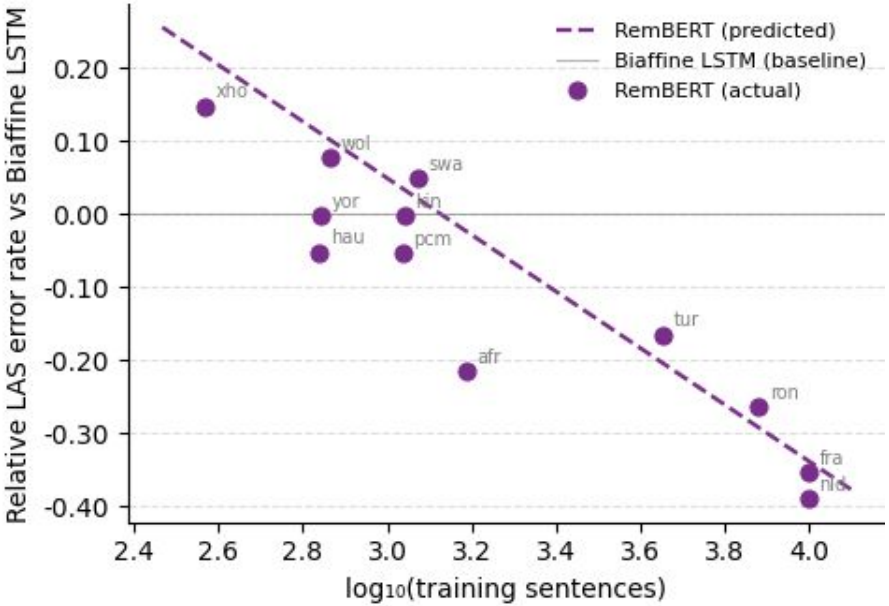
- Not all improvements are equally impressive
  - 60% to 62%
  - 97% to 99%
- Take Biaffine LSTM as baseline

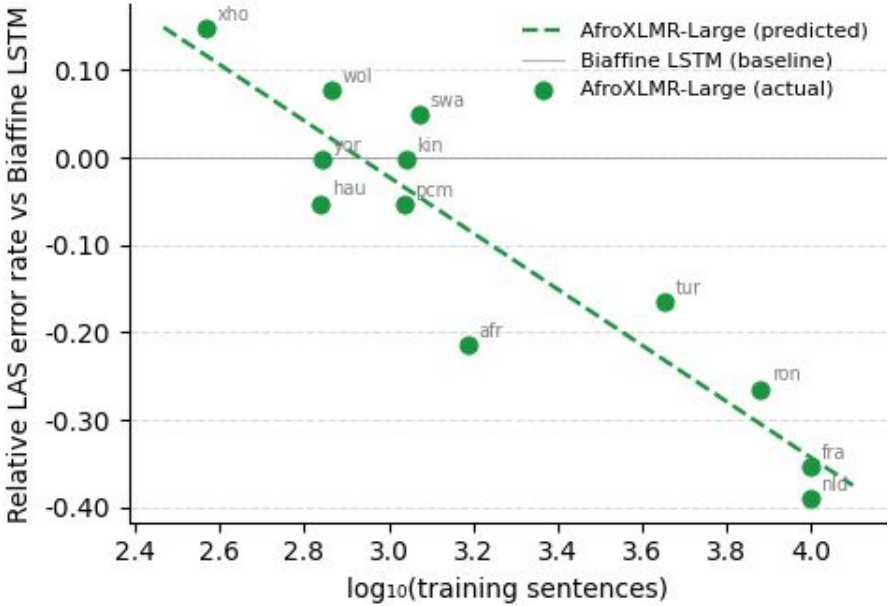
$$RER_{\text{LAS}}(\ell) = \frac{\text{LAS}_{\text{Biaffine LSTM}}(\ell) - \text{LAS}_{\text{TF}}(\ell)}{100 - \text{LAS}_{\text{Biaffine LSTM}}(\ell)}$$

Results


Relative LAS error rate
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
StackPointer
AfroXLMR-Large
RemBERT
Biaffine LSTM (baseline)
xho (369)
hau (690)
yor (694)
wol (736)
pcm (1,085)
kin (1,098)
swa (1,187)
afr (1,547)
tur (4,508)
ron (7,619)
nld (10k)
fra (10k)
Language (training sentences)

Linear Mixed Models





Crossover Point

Model	Metric	Sentences
AfroXLMR-large	LAS	854
RemBERT		1,339
AfroXLMR-large	UAS	839
RemBERT		1,311

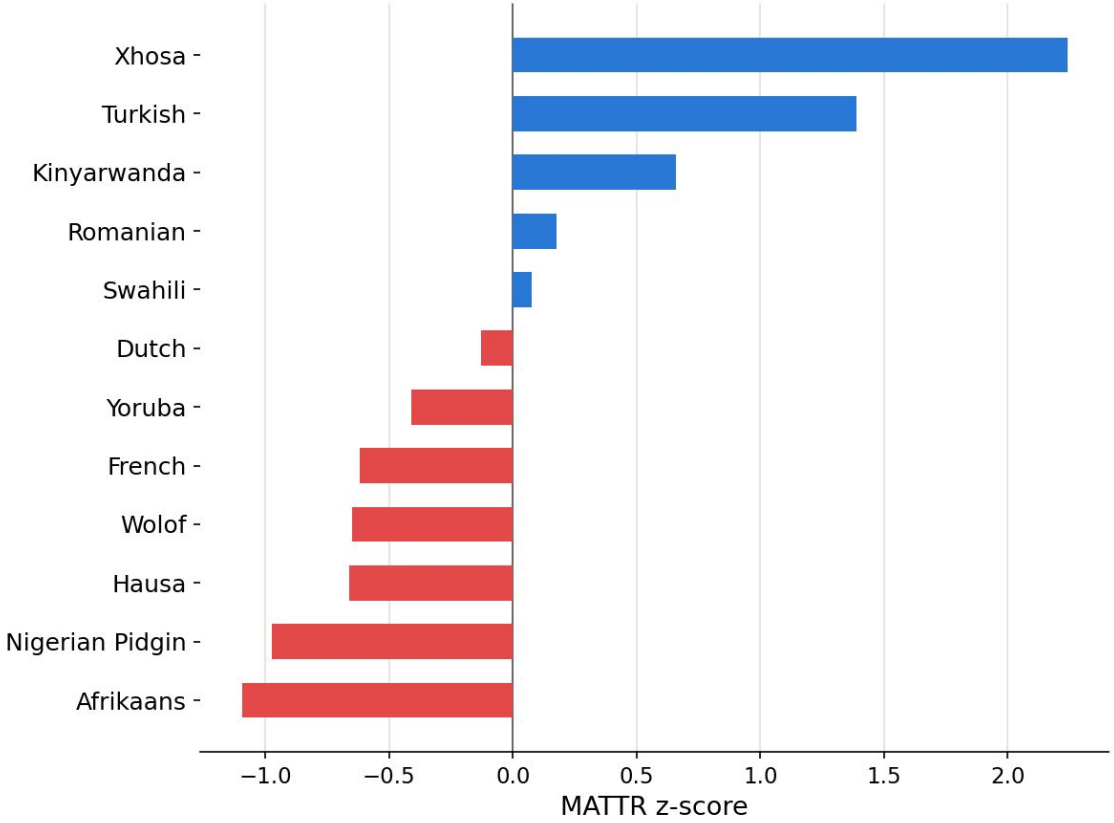
Morphological Complexity

Q: Does morphological complexity affect the relative performance of parsers beyond training-data size?

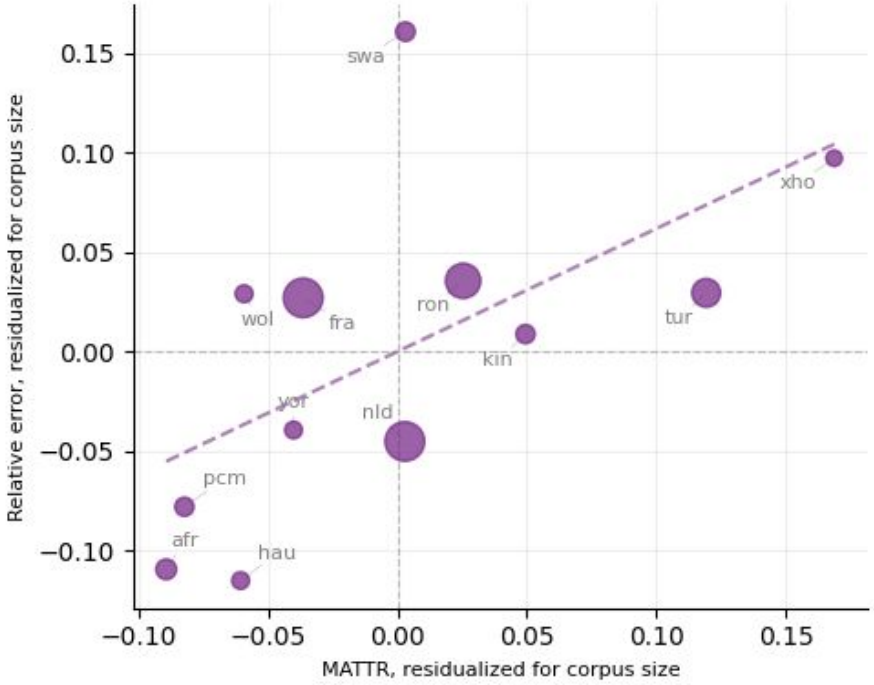
Moving-Average Type-Token Ratio

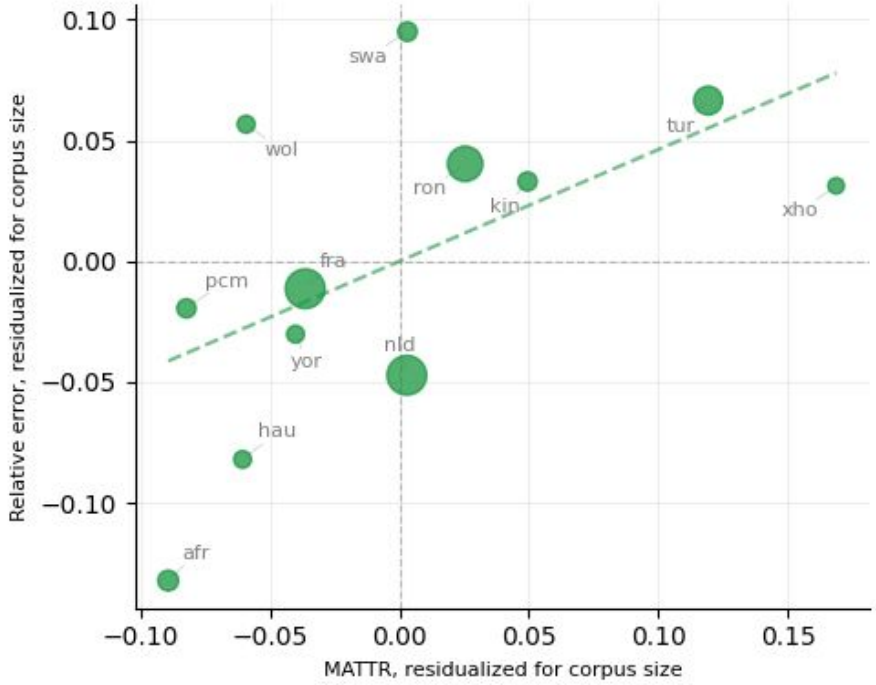
$$\mathrm{MATTR}(w) = \frac{1}{N - w + 1} \sum_{i=1}^{N-w+1} \frac{|\{t : t \in W_i\}|}{w}$$

MATTR


Xhosa
Turkish
Kinyarwanda
Romanian
Swahili
Dutch
Yoruba
French
Wolof
Hausa
Nigerian Pidgin
Afrikaans
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
MATTR z-score

Linear Mixed Models





Limitations and Future Work

Limitations

- Word-embedding quality is a confound for non-transformer models
- Typological differences can't be fully separated from resource level
- Small test sets for some languages limits statistical precision

Future Work

- Other NLP tasks
- More morphologically rich languages
- Incorporating prosodic information
- Controlled subsampling
- Better measures of morphological complexity

Conclusion

- LSTM-based parsers are competitive and often better in low-resource settings
- Transformers increasingly outperform LSTMs as training data increases
- Morphological complexity helps explain variation in the performance gap

Acknowledgements

- Janet Pierrehumbert for inspiring our central hypothesis
- The Princeton Language and Intelligence (PLI) Seed Grant Program for their support of our work
- Princeton Linguistics Department for supporting our conference participation